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The Role of Ultra-Deep Resistivity Tools in Detecting Oil-Water-Contact in a Faulted Regime While Geosteering

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Abstract

Geosteering horizontal wells in faulted carbonate reservoirs presents significant operational challenges, particularly in detecting variations in fluid contacts such as the oil-water contact (OWC). These variations are often influenced by complex fault systems, compartmentalized reservoir blocks, and historical production and water injection, which can displace fluid contacts and lead to undulating boundaries. Conventional resistivity-based Logging While Drilling (LWD) tools frequently lack the depth of investigation necessary to detect these contacts early, increasing the risk of water breakthrough and suboptimal well placement.

This paper demonstrates the application of ultra-deep azimuthal resistivity (UDR) tools in accurately identifying OWC variations while geosteering in a structurally complex, mature carbonate dome structure. By integrating ultra-deep resistivity measurements—capable of imaging tens of meters from the wellbore—with real-time inversion algorithms and structural models, the geosteering team successfully identified bed boundary undulations, fault cuts, and fluid contact shifts during drilling.

Operational results from this case confirm that the integration of UDR data with pre-existing models and real-time workflows significantly reduced subsurface uncertainty, improved wellbore positioning, and enhanced reservoir exposure while avoiding aquifer contact. The high-resolution inversion models enabled early detection of OWCs and compartment boundaries, providing valuable inputs for optimized well trajectories and future development planning.

Introduction

Geosteering horizontal wells in complex carbonate reservoirs is a critical operation aimed at maximizing hydrocarbon recovery and minimizing non-productive time. In faulted and mature fields, maintaining well placement within the optimal reservoir interval is particularly challenging due to structural discontinuities, displaced fluid contacts, and lateral variations in formation properties (Powers et al. 1966; Steineke et al. 1958). These subsurface complexities increase the risk of premature aquifer breakthrough, high water cut, and reduced reservoir exposure, all of which can compromise well productivity and asset value.

A key operational challenge in such environments is the precise real-time identification of fluid interfaces, particularly oil-water contacts (OWCs), during drilling operations. These OWCs are typically

non-planar and subject to vertical or horizontal displacement due to tectonic faulting, stratigraphic variations, or pressure differentials induced by water injection and production drawdown (Taher et al., 2024). Conventional Logging While Drilling (LWD) resistivity tools typically investigate a limited volume around the borehole, restricting their ability to detect undulating fluid contacts or remote bed boundaries until the wellbore is in close proximity.

Recent advancements in ultra-deep azimuthal resistivity (UDR) technology have introduced a step-change in formation evaluation and geosteering capability. These tools can image formation resistivity tens of meters away from the borehole, enabling early identification of fluid contacts, bed boundaries, and fault-induced compartmentalization in real time (Antonsen et al. 2018). When combined with inversion modeling workflows, UDR data provides a detailed resistivity profile of the surrounding formation, offering significant operational advantages in geosteering decision-making and reservoir contact optimization.

This paper presents a case study demonstrating the application of UDR tools in a horizontal well drilled through a structurally complex, mature carbonate dome structure. It highlights how integrating UDR measurements with real-time inversion workflows and structural models improved wellbore positioning, enabled proactive trajectory corrections, and minimized drilling risks associated with water production in a challenging subsurface environment.

Statement of Theory and Definitions

In faulted carbonate reservoirs, structural complexity and prolonged production histories often lead to displaced and irregular oil-water contacts (OWCs). These fluid interfaces exhibit vertical and lateral heterogeneity due to fault offsets, compartmentalization, and pressure differentials induced by water injection and production activities. (Taher et al., 2024; Powers et al. 1966). Managing these variations is essential to optimize well placement and reduce water production risk during horizontal drilling.

Traditional formation evaluation techniques, such as pressure testing and conventional LWD resistivity, are limited by shallow depth of investigation and vertical resolution. Ultra-deep azimuthal resistivity (UDR) tools address this limitation by acquiring multi-spacing, multi-azimuth electromagnetic measurements tens of meters from the borehole (Antonsen et al. 2018). These measurements identify variations in formation resistivity indicative of bed boundaries, fluid contacts, and structural features ahead of the bit.

The acquired data is processed through inversion algorithms that convert raw measurements into resistivity profiles. Deterministic inversion provides a single best-fit model, while stochastic inversion generates multiple possible subsurface scenarios (Dong et al. 2015). Recent advancements introduced UDR tools with nine electromagnetic components, enabling full three-dimensional inversion imaging of the surrounding formation (Elkhamry et al. 2024). This enhances the detection of fluid contacts and structural discontinuities in all directions around the wellbore, supporting real-time trajectory decisions in complex reservoirs.

Description and Application of Equipment and Processes

To address the challenge of accurately detecting fluid contacts and structural discontinuities in a faulted carbonate reservoir, this study deployed ultra-deep azimuthal resistivity (UDR) tools in a horizontal well drilled through a complex carbonate dome structure. These advanced Logging While Drilling (LWD) tools use multiple transmitter-receiver spacings and azimuthal electromagnetic measurements to acquire resistivity data at radial depths of investigation reaching up to 68 meters from the borehole.

Ultra-deep resistivity measurements are based on electromagnetic wave propagation principles, where differences in formation resistivity are detected as variations in phase shift and attenuation between transmitted and received signals. By deploying multi-component, multi-spacing arrays, these tools generate a composite dataset that captures formation resistivity at various depths and azimuthal positions relative to

the borehole, making them highly effective for mapping bed boundaries, fluid contacts, and fault cuts at substantial distances from the wellbore (Antonsen et al. 2018).

The raw tool responses are processed through real-time inversion algorithms to convert the measured electromagnetic signals into resistivity profiles surrounding the borehole. Two primary inversion approaches are typically applied: deterministic and stochastic. Deterministic inversion produces a single, best-fit resistivity model based on the input data and known formation parameters, whereas stochastic inversion generates a distribution of possible models by randomly sampling formation properties within predefined constraints (Dong et al. 2015). Stochastic inversion methods, such as Markov Chain Monte Carlo simulations, offer a more comprehensive assessment of subsurface uncertainty by exploring a wide range of possible resistivity distributions.

In this application, unconstrained inversion techniques were preferred to avoid introducing potential bias into the resistivity models. While unconstrained inversions demand higher computational resources and longer processing times, they provide objective imaging outcomes free from operator-imposed assumptions, making them particularly valuable in structurally complex environments where limited offset data is available.

The resistivity inversion results were integrated with pre-existing 3D seismic models, offset well data, and real-time drilling parameters within a geosteering workflow. This integration enabled the identification of OWC variations, bed boundary undulations, and fault displacements during drilling, allowing for timely trajectory adjustments to avoid premature aquifer breakthrough and maximize hydrocarbon exposure.

Additionally, signal processing techniques based on azimuthal antenna coupling and phase/attenuation measurements were applied to enhance detection accuracy at greater distances. The measurements were acquired at multiple transmitter-receiver spacings and azimuthal positions around the borehole, producing a high-resolution, three-dimensional resistivity distribution of the formation (Antonsen et al. 2018; Dong et al. 2015).

The application of ultra-deep resistivity tools, combined with inversion modeling and structural interpretation, represents a significant advancement in geosteering technology for mature, faulted reservoirs with complex fluid distributions. This approach provided valuable real-time subsurface insights that improved wellbore placement, reduced operational risks, and enhanced reservoir management strategies.

Presentation of Data and Results

A horizontal well, designated KK, was drilled through a mature carbonate reservoir in a dome structure to evaluate the operational effectiveness of ultra-deep resistivity (UDR) tools in a faulted and structurally complex environment. The field, characterized by a long production history and extensive water injection, presented multiple challenges including displaced and undulating oil-water contacts (OWCs), complex resistivity distributions, and lateral variations in lithology.

Prior to drilling, a comprehensive reservoir characterization study integrated 3D seismic interpretation, fault mapping, and available offset well data. This revealed significant structural discontinuities and compartmentalization within the reservoir, with variable OWCs anticipated across fault blocks. The operational objective was to place the well above the OWC, maximizing reservoir exposure while avoiding premature water contact.

During drilling, the ultra-deep resistivity tool demonstrated the capability to detect formation boundaries and fluid contacts up to 68 meters from the borehole. Early in the horizontal section, inversion imaging revealed a pronounced resistivity contrast consistent with the OWC several meters below the wellbore. This real-time data allowed the geosteering team to confirm the well's position relative to the OWC and maintain the trajectory within the optimal pay zone.

As the well progressed, the UDR tool detected several resistivity anomalies indicative of bed boundary undulations and potential fault cuts. These anomalies were corroborated by corresponding structural features

identified in the pre-drill seismic model. In one instance, an abrupt shift in resistivity inversion results suggested a fault-induced compartment boundary, prompting an inclination adjustment to maintain well placement within the productive interval.

The resistivity inversion models also revealed lateral variations in fluid contacts consistent with the anticipated effects of long-term water injection. Water breakthrough zones were identified as low-resistivity anomalies approaching the wellbore in certain fault blocks, providing early warning of production risks. In these areas, geosteering decisions were made proactively to maintain well placement above the moving contact.

The combination of ultra-deep resistivity measurements and real-time inversion imaging significantly reduced subsurface uncertainty during drilling. Compared to conventional resistivity-based LWD tools, which typically investigate only a few meters from the borehole, the UDR system enabled early detection of distant OWCs and structural features, improving overall well placement accuracy and minimizing non-productive time.

Operational outcomes from this application included increased net reservoir exposure, improved trajectory control in faulted zones, and reduced risk of water production. The final well path successfully maintained placement above the variable OWCs throughout the lateral section, validating the integration of ultra-deep resistivity data with inversion workflows and structural interpretation models as a reliable geosteering strategy for mature, faulted carbonate fields.

Presentation of Data and Results

A horizontal well, referred to as KK, was drilled across a faulted carbonate reservoir within the Dome structure. The reservoir is characterized by a long production history, extensive water injection, and complex structural compartmentalization that displaced the original oil-water contact (OOWC) and introduced variable oil-water contact (OWC) levels across fault blocks. The primary operational objective was to position the well within the optimal reservoir interval, maximizing reservoir exposure while avoiding premature contact with the underlying aquifer.

Prior to drilling, a comprehensive pre-job study was conducted integrating 3D seismic interpretation, structural mapping, and available offset well data. This subsurface model indicated the presence of multiple faults with varying throws, lateral facies changes, and potential undulating OWCs within the reservoir interval. These structural uncertainties posed a significant geosteering challenge, particularly in regions of limited historical well control.

Ultra-deep azimuthal resistivity (UDR) tools were deployed to enhance fluid contact detection and structural imaging while drilling. The tools demonstrated the capability to detect formation boundaries and resistivity contrasts up to 68 meters from the wellbore. Early in the lateral section, inversion images revealed a clear resistivity boundary consistent with the OWC several meters below the borehole, confirming the well's placement within the pay zone (Figure 1). The geosteering team was able to maintain this position by making real-time trajectory adjustments based on inversion data.

As drilling progressed, the UDR tool identified several high-contrast resistivity anomalies interpreted as bed boundary undulations and fault displacements. These anomalies aligned well with fault features anticipated from the pre-drill seismic model. One notable inversion result indicated an abrupt resistivity shift suggestive of a fault throw, prompting an immediate inclination adjustment to maintain well placement within the productive interval.

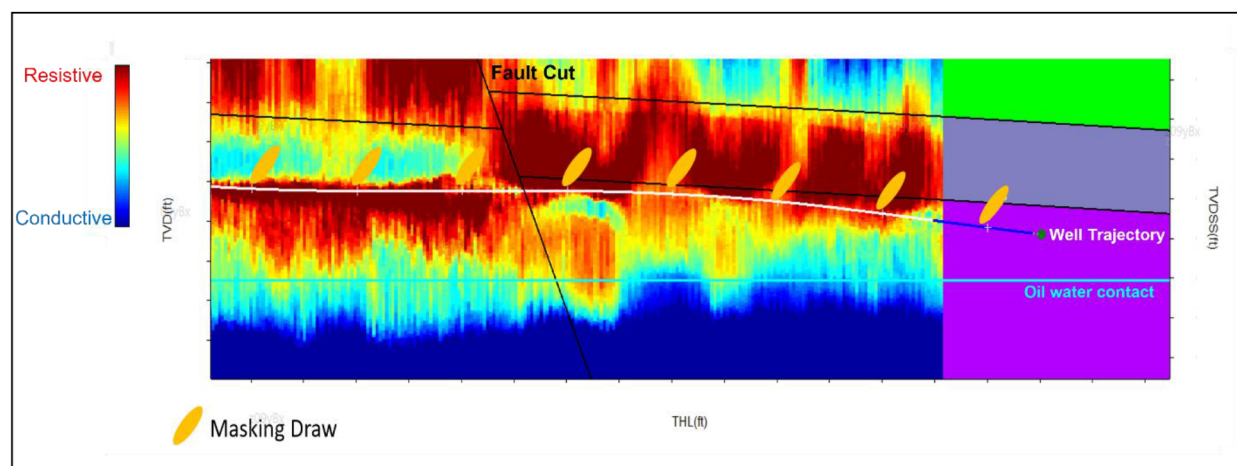


Figure 1—Inversion imaging output from the ultra-deep azimuthal resistivity tool showing the oil-water contact (OWC) several meters below the borehole early in the horizontal section.

The inversion results also highlighted lateral variations in fluid contacts between fault blocks. Zones of reduced resistivity approaching the wellbore were interpreted as displaced water-up-to (WUT) zones, associated with historical water injection and aquifer movement. The ability to detect these anomalies several tens of meters ahead of the bit provided early warning, enabling proactive wellbore steering decisions to avoid water-prone sections.

In total, the application of ultra-deep resistivity tools and inversion modeling in this well delivered significant operational benefits, including:

- Improved trajectory control in structurally complex zones.
- Consistent real-time identification of OWCs and bed boundaries.
- Early detection of aquifer breakthrough risks.
- Maximized net pay exposure and reduced drilling non-productive time.

These results confirmed that ultra-deep resistivity imaging, when combined with inversion workflows and structural interpretation, substantially enhances well placement accuracy and risk management in faulted, mature carbonate reservoirs.

Conclusions

The application of ultra-deep azimuthal resistivity (UDR) tools in a faulted, mature carbonate reservoir within the dome structure successfully demonstrated the technology's operational and subsurface imaging value. The integration of UDR measurements with inversion workflows and geological models significantly reduced geosteering uncertainty and improved wellbore positioning in a structurally complex environment.

The study confirmed that UDR tools enable early detection of oil-water contacts (OWCs) and bed boundary undulations at distances far beyond the capability of conventional resistivity tools. The high-resolution inversion models allowed geosteering teams to proactively manage trajectory changes in response to approaching aquifer contacts and fault-induced compartment boundaries.

Operational outcomes from this case included:

- Enhanced net reservoir exposure and optimized horizontal well trajectory.
- Reduced risk of water production and premature aquifer breakthrough.

- Improved subsurface understanding through the detection of lateral fluid contact variations and compartmentalization effects.
- Early identification of resistivity anomalies indicative of fault displacements and water movement.

Overall, the integration of ultra-deep resistivity imaging technology with real-time inversion modeling represents a substantial advancement in geosteering workflows for mature, structurally complex carbonate reservoirs, offering new opportunities to maximize hydrocarbon recovery and optimize field development strategies.

Novel/Additive Information

This paper introduces a novel operational integration of ultra-deep azimuthal resistivity (UDR) measurements with inversion-derived subsurface models to improve geosteering decision-making in faulted, compartmentalized carbonate reservoirs. Unlike conventional LWD tools with limited depth of investigation, UDR systems provide real-time, azimuthal resistivity imaging at distances exceeding 60 meters from the borehole, enabling early detection of formation boundaries, fluid contacts, and fault displacements.

The study's application in a dome represents a field-scale demonstration of this capability in a structurally complex, mature field environment. It highlights the technology's ability to resolve displaced oil-water contacts, identify water-up-to (WUT) zones, and improve well placement precision by integrating inversion results with pre-drill structural interpretations and real-time drilling data.

This operational workflow advancement contributes a practical, field-proven solution for managing fluid distribution complexity in faulted reservoirs and enhancing subsurface imaging capability during horizontal drilling campaigns. The demonstrated improvements in wellbore placement accuracy, reduced drilling risk, and optimized reservoir exposure illustrate the significant operational value of UDR technology for similar mature, structurally complex fields worldwide.

Nomenclature

Symbol/Abbreviation	Definition
OWC	Oil-Water Contact
OOWC	Original Oil-Water Contact
GOC	Gas-Oil Contact
WUT	Water-Up-To zone — the highest vertical point in a compartment where water is present
UDR	Ultra-Deep Azimuthal Resistivity
LWD	Logging While Drilling
DOD	Depth of Detection — maximum radial depth from the borehole measurable by a given resistivity tool
Inversion	A mathematical process of converting raw geophysical measurements into a modeled subsurface resistivity image
Deterministic Inversion	Inversion method providing a single, most-likely resistivity model based on measured data
Stochastic Inversion	Inversion method generating a distribution of possible resistivity models based on random sampling of formation properties
Seismic Acoustic Impedance (AI)	Product of rock density and seismic velocity, used to detect fluid or lithological contrasts

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